Week 10 Assignment

Multiple regression and ANCOVA

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf`, along with your Data Safari submission, via Canvas before Week 11's lecture (Monday). See the syllabus for the late policy.

Note: This week also includes a Data Safari exercise — a separate, short assignment. Both are due at the same time.

Part 1: Multiple regression

Using the built-in `trees` dataset:

1. Fit `lm(Volume ~ Girth + Height)`.
2. Compare its R^2 to the simple `Volume ~ Girth` model from Week 9. Support your answer with `anova()`.

Did adding `Height` meaningfully improve the fit?

Part 2: ANCOVA

Using this week's tutorial's simulated `ancova_data` (or re-simulate it with a different `set.seed()` of your choosing):

1. Fit `lm(final_height ~ start_height + treatment)`.
2. Fit the version with an interaction term and compare the two models.
3. In 2–3 sentences: explain what the `treatment` coefficient means in the no-interaction model, and what a significant interaction would have meant instead.